PROJECT NEXUS: ARCHITECTURE DECISION RECORD

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1. Problem Statement & Legacy Flaws

Legacy DynamoDB implementation suffered from eventual consistency reconciliation anomalies during high traffic bursts, resulting in $42,000 ledger drift. A strictly ACID-compliant transactional persistence engine is required.

2. Approved Decision: PostgreSQL with TimescaleDB

We formally adopt PostgreSQL 16 with TimescaleDB hypertables for all double-entry ledger accounts. Rationale: Strict ACID transactions, native chunked audit queries under 650ms, and compliance with European Central Bank DORA directives.

3. Lead Owner & Delivery Milestone

Lead Architect: Elena Vance. Target Sprint: Sprint 43 (October 24, 2026). Status: Approved and in active execution.

4. External Blockers & Third-Party Dependencies

Critical dependency: Core Banking Gateway API v3.4 by FinVance Inc. Status: UNRESOLVED. Vendor support ticket #FV-9021 remains open with no SLA commitment or staging endpoint delivery date.

5. Cryptographic Key Management

Decision recorded to rotate AWS KMS customer-managed signing keys on a 14-day cycle. (Note: No rationale or compliance justification was provided for this 14-day frequency; standard security policy specifies 90 days).